All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented. If you use an online resource to solve any problem, please appropriately cite that source.

Notes:

a. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.

b. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.

c. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).

Note that we use the notation $v = (v_1, v_2, \dots, v_n)$ and equivalently,

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

Problem 1. (Inner Product Properties.) Consider the following vectors and scalars:

$$v = (v_1, v_2, \dots, v_n),\ w = (w_1, w_2, \dots, w_n),\ u = (u_1, u_2, \dots, u_n), \qquad \alpha, \gamma$$

Verify the following properties. That is, show that the right-hand side of the expression is equal to the left-hand side.

a. $(\gamma v)^\top w = \gamma(v^\top w)$

b. $(u + v)^\top w = u^\top w + v^\top w$

Problem 2. (Inner Products.) Consider scalars v_i for $i = 1, \dots, n$.

a. Write $\sum_{i=1}^n v_i^2$ in terms of an inner product of two vectors.

b. Write the average of the scalars as the inner product of two vectors—i.e., $\frac{1}{n}\sum_{i=1}^n v_i$

c. For n even, write $\sum_{i=1}^n v_i$ as the inner product of two vectors where one of the vectors is $w = (v_1, 0, v_2, 0, \dots, v_n, 0)$. What is the dimension of w?

Problem 3. (Subspaces and dimension.) A nonempty subset $\mathcal{S}$ of $\mathbb{R}^n$ is a *subspace* if, for any scalars α, β, we have that

$$x, y \in \mathcal{S} \implies \alpha x + \beta y \in \mathcal{S}.$$

Consider the set $\mathcal{S}$ of points in $\mathbb{R}^3$ such that

$$\mathcal{S} = \{(x_1, x_2, x_3) \mid x_1 + 2x_2 + 3x_3 = 0,\ 3x_1 + 2x_2 + x_3 = 0\}$$

Show that $\mathcal{S}$ is a subspace of $\mathbb{R}^3$. What is the dimension of $\mathcal{S}$?

Problem 4. (Linear Maps.) A map $T : \mathbb{R}^n \mapsto \mathbb{R}^m$ is a linear transformation if $T(u+v) = T(u)+T(v)$ and $T(cu) = cT(u)$ for vectors u, v, w and scalar c. Consider the map $T : \mathbb{R}^4 \to \mathbb{R}^4$ that takes a vector in $\mathbb{R}^4$ and reverses its entries—i.e.,

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \mapsto \begin{bmatrix} v_4 \\ v_3 \\ v_2 \\ v_1 \end{bmatrix}$$

a. Show this map is a linear transformation.

b. Find a matrix representation of this map—i.e., find a matrix A such that $T(u) = Au$. **Hint**: use the standard basis vectors and apply T to each of those.

Problem 5. (Matrix-vector multiplication.) Suppose x is an n-vector and A an $n \times n$ matrix. Write the expression $x^\top Ax$ in terms of the vector entries x_i and matrix entries A_{ij}. Does the expression simplify if the matrix A is diagonal? How?

Problem 6. (VMLS 2.8, Integral and derivative of polynomial.) Suppose the n-vector c gives the coefficients of a polynomial $p(x) = \theta_0 + \theta_1 x + \cdots + \theta_{n-1}x^{n-1}$.

a. Let a and b be scalars. Show that $p(ax + by) = ap(x) + bp(y)$ for all $x, y \in \mathbb{R}$. This means that a polynomial is a linear transformation.

b. Let a and b be scalars. Find an n-vector v for which $v^\top \theta = \int_a^b p(x)dx$. This means that the integral of a polynomial over an interval is a linear function of its coefficients.

c. Let a be a scalar (real number). Find an n-vector v for which $v^\top \theta = p'(a) = \frac{dp}{dx}(a)$. This means that the derivative of the polynomial is a linear transformation.

Problem 7. (VMLS 3.9, Difference of squared distances..) Determine whether the difference of the squared distances to two fixed vectors c and d, defined as

$$f(x) = \|x - c\|_2^2 - \|x - d\|_2^2$$

is linear, affine, or neither. If it is linear, give its inner product representation, i.e., an n-vector v such that $f(x) = v^\top x$. If its affine, give its inner product representation, i.e., an n-vector v and scalar b such that $f(x) = v^\top x + b$. If it is neither linear nor affine, give specific x, y, α, and β for which superposition fails, i.e.,

$$f(\alpha x + \beta y) \neq \alpha f(x) + \beta f(y)$$

(Provided $\alpha + \beta = 1$, this shows the function is neither linear nor affine.)

Problem 8. (VMLS 3.24, Distance versus angle nearest neighbor..) Suppose that $z_1, z_2, \ldots, z_m$ is a collection of n vectors, and x is another n-vector. The vector z_i is a *(distance) nearest neighbor* of x if it is the vector in the collection that is closest to x—i.e.,

$$\|x - z_i\|_2 \leq \|x - z_j\|_2 \ \forall \ j = 1, \ldots, m.$$

We say z_i is the angle nearest neighbor of x if it is the vector in the collection that makes the smallest angle with x—i.e.,

$$\angle(x, z_i) \leq \angle(x, z_j) \ \forall \ j = 1, \ldots, m.$$

a. Give a simple specific numerical example where the (distance) nearest neighbor is not the same as the angle nearest neighbor.

b. Now suppose that the vectors $z_1, z_2, \ldots, z_m$ are normalized which means that $\|z_i\|_2 = 1$ for all $i = 1, \ldots, m$. Show that in this case the distance nearest neighbor and the angle nearest neighbor are always the same. Hind: You can use the fact that the arccos is a decreasing function, i.e., for any u and v with $-1 \leq u < v \leq 1$, we have $\arccos(u) > \arccos(v)$.